Museum Guide Robot Initial Prototype Summary

Objective

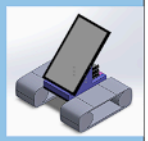
Create a guide robot that helps visitors tour a museum

Criteria:

- Has Intuitive user-interface
- Meets needs of different users
- Has multilingual support
- Provides accessibility options

Design

Current Model: Tank drive base with 3D printed mount for electronics and tablet



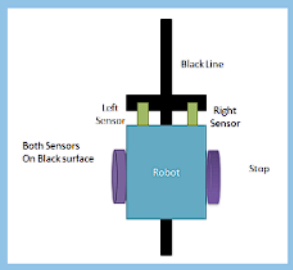
Features:

- Full mobility and can turn in place
- Multiple sensors to detect path
- Multiple arduinos to connect sensors, motors and tablet
- Bluetooth connection to tablet

Testing

Criteria

- Test robot's ability to navigate to exhibits
- Test robot's ability to stop when presented with obstacle
- Test communication between tablet/web app and robot



Lessons Learned

- Flesh out project design more before we actually start
- Spend more time picking out parts that are compatible with each other that meet our needs
- Do more testing in real-world environments

If we had more time…

- We would flesh out our obstacle avoidance system to be more accurate - maybe use anticipatory system
- We would make our user-interface more friendly and accessible to different people

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Summary

Museums are hubs of artistic expression that allow people to admire famous art pieces and reflect on their lives and historical events. However, they also tend to get overcrowded with visitors due to their popularity. There are also issues of tours not being personalized to fit specific people's needs. For example, tour guides may not speak the visitor's language, and they are likely at a different level of understanding than the visitors themselves so it can be difficult for visitors to enjoy the tour properly. To solve this problem, our team proposes the Museum Guide Bot which leads visitors to their exhibit of choice, all while giving them information about each exhibit in their preferred language and style.

To help us create the best possible design for our robot, we conducted interviews with potential stakeholders. Zulfeqar Ali, an automotive engineer with over a decade of experience in the industry and a Master's degree in Computer Engineering, told us to include an infrared sensor. This would help us detect obstacles and prevent any unwanted collisions in the museum. In addition, Catie Cuan, an expert roboticist with a Master's and PhD from Stanford in Mechanical Engineering and Computer Science, informed us that people are more likely to approach our robot if it has smoother, less abrupt movements. It would help it seem more human-like and therefore more approachable.

To better understand industry standards for social navigation, we reviewed some academic articles. One article, "A Study in Zucker: Insights on Interactions between Humans and Small Service Robots" by Alex Day and Ioannis Karamouzas, explained different social navigation robots and the benefits and downsides of each. From this article, we learned that we should use a non-anticipatory approach, meaning that the robot follows a set path. This would not only avoid the unnecessary need for more complex code and hardware but most importantly improve user safety.

To best serve visitors, we decided that our initial design of the Museum Guide Bot will have three main aspects. It will need to have a friendly user interface, it needs to be able to follow lines to the corresponding exhibits, and it needs to be able to detect objects or people and stop before hitting them.

One core feature of the Museum Guide Bot is the ability to navigate to exhibits. To achieve this goal, we implemented a line-following system using three infrared sensors that differentiate between the colors of the line and the ground. These three sensors will communicate with each other to direct the robot forward, left, and right until it reaches its destination.

Another core feature of the robot is visitor safety. We need to make sure that no visitors are hurt while the robot moves around and goes to different exhibits. To ensure that visitors are safe, we implemented an object detection algorithm. After consulting our stakeholders, we found that an

infrared sensor is best for this operation, and we decided to place the sensor at the front of the robot since that is the direction the robot will be moving in. We also found that the most ideal distance for the sensor to detect an object and signal the robot to stop was around 1 foot.

The Museum Guide Bot can tell visitors information about the various exhibits. We decided that the most effective route was to use information written by the tour guides themselves. We would also use any information that the museum itself has about each art piece and exhibition because each museum has its unique exhibits depending on what art pieces they are showcasing. This would remove any loss of information between human and robot tours. Lastly, this feature communicates with the line-following system so that the robot knows which exhibit to go to and whether or not to repeat any information about each art piece.

With these basic features, we believe we were able to make a successful prototype. However, there are still multiple limitations that we would work on and improve. Some limitations are a somewhat unintuitive user interface, object detection that needs better accuracy, and improved coding for line following and smoother robot movement.

Introduction

As inflation continues to rise and the cost of living increases year after year, the number of museum workers and tour guides dwindles as they search for better jobs with more opportunities and higher pay [1]. Due to this, museums are understaffed nationwide, and finding tour guides to accommodate every visitor at museums is becoming harder and harder. On top of this, museums can get high traffic, so the understaffing issue is exacerbated and visitors aren't being properly provided for.

In terms of the actual tour guides themselves, it becomes difficult for those with disabilities or foreigners who are visiting to get a great experience with a human tour guide. If museums want to accommodate guides for the disabled, they would have to pour extra resources into areas such as audio guide systems, text-based guides, and more, making it inefficient for the museum itself. For those who don't speak the native language of the country, the museums would have to hire additional staff who are multilingual and can communicate effectively with foreigners. This causes more spending from the museum and faces unnecessary costs. When a museum becomes crowded during peak hours, it becomes extremely difficult to hear the tour guides themselves due to increased volume. It becomes even harder for the museum guides themselves when there's a large group all trying to join one tour, causing increased traffic in an already congested area.

To solve both of these issues, we created a museum guide robot that aims to improve the staff shortage at museums and further improve the visitor experience by providing more accessibility options for those with disabilities and therefore a more personalized visit. This robot would also

save a lot of money for museums as there is no wage being paid to the robots. This lets the museum be able to purchase as many as they need depending on how many visitors they attract. The robot can also easily accommodate those with disabilities, by combining audio and text-based guides all into one. This makes it easy for those who are visually or audibly impaired to enjoy the same experience as everyone else.

It is important to note that we built a smaller desktop version of our original idea as an initial proof of concept before fully fleshing our idea out. Our original idea would likely be around 4-5 feet tall so that visitors could easily interact with it. However, the version we built is only around 1 foot tall with less sophisticated functionality and movements than what the full-sized robot would have.

The rest of this report will focus on the design and features of our robot, including how our research led us to improve and edit our initial design.

Initial Design

Our team's initial design consisted of a large human sized robot as seen in **Figure 1.** This initial design had various language and accessibility capabilities which would allow for all users to have a fulfilling museum experience.

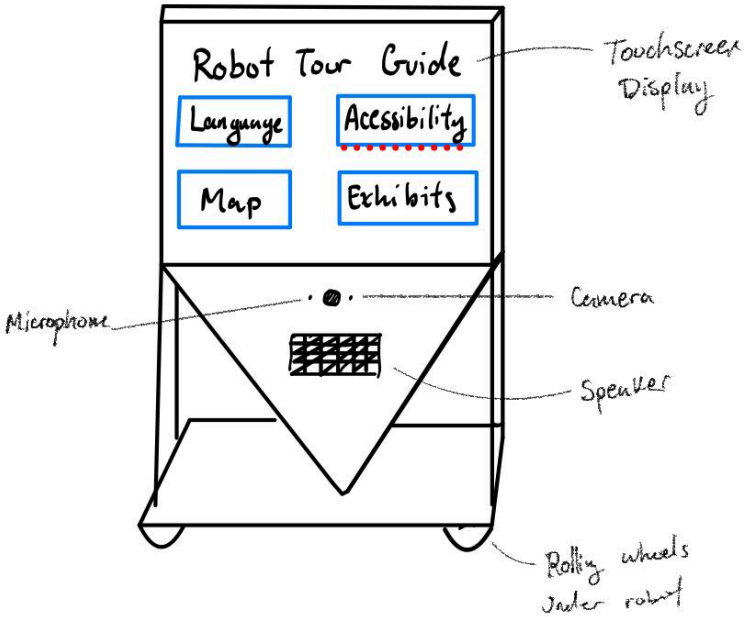

Figure 1: A model image of our initial museum guide robot design
Credit: Haotian Li

However, due to the limited scope of a first iteration prototype, our team chose to revise our design and focus on creating a table-top demo with limited functionality and features. This demo would focus on exhibiting the guide robot's ability to navigate to mock exhibits and provide information about them. We believed that a simple demonstration of navigation and informative capabilities would suffice and showcase the possibility of other applications of our robot. This demo robot consisted of infrared (IR) sensors to assist with line following and object detection. This would be plugged into an Arduino which was running a script to detect the various colors of the lines, utilizing it to see if the robot was staying on track. To easily send exhibit information to the user, we connected the robot to a tablet via a Bluetooth module, reducing the amount of wiring needed. We also used a terminal app, allowing the user to easily send and receive information from the Arduino. We placed all of this on a base with tank treads. We decided on tank tread wheels specifically as it would increase the robot's ability to maneuver in tight corners and assist in keeping the robot on the line as it navigates to its destination.

Research

In order to create the best possible design for our museum guide robot, we conducted research to find what potential needs our future users and museum tour guides would have when using our robot. We also interviewed technical experts in order to find the best devices for things such as sensing people and objects and how we should make our robot as user-friendly as possible, making it approachable and easy to interact with.

Interviews

Technical Consultation with Zulfeqar Ali

While considering the potential risks of our robot running into an obstacle or visitor, we interviewed Zulfeqar Ali, an automotive engineer with over a decade of experience in the industry and a Master's degree in Computer Engineering. We knew that we would need some type of sensing system so that the robot could follow a line to go around the museum while also being able to prevent hitting obstacles and people. However, we weren't sure what type of sensors were best to use for our circumstances, which is why we went to Ali for help, who has experience with the use of sensors, especially in vehicles.

After telling Ali the scope of our project and our aim of making the robot stop automatically if someone or something comes in front of it, he recommended that we use an infrared sensor [2]. This is best for object detection because all we need to do is sense if there is some object or person in front of the robot. It does not need to identify what the object is, so devices like a color sensor or normal camera would not be necessary and would only result in additional hassles that wouldn't give an advantage over a traditional infrared sensor. Additionally, Ali suggested implementing some type of speaker system to alert anyone who might be in the way of the robot to step aside [2]. Taking Ali's recommendations into account, we decided to implement an

infrared sensor that could detect light up to around 30 inches away and updated our design to try and fit a speaker system.

Interview with Robotics expert Catie Cuan

In addition to the safety of visitors, how the visitors and our robot interact is also very important. The most fundamental job of our robot is to lead visitors to different exhibits and inform them about art pieces, which entails lots of interaction between visitors and the Museum Guide Robot, so we had to make sure that visitors will have a good experience when using the robot while going through our design process. That is where we went to an interview with Catie Cuan, an expert roboticist with a Master's and PhD from Stanford in Mechanical Engineering and Computer Science. When explaining how robots can seem more natural and friendly to humans, Cuan talked about nonverbal communication and how "you can express so much through simply the way that your body moves" [3]. This is important in the case of our project because robots often perform "utilitarian" movements like sudden turns, starts, and stops which make people uncomfortable and more apprehensive towards robots. Using this information, we decided that the infrared sensor which we implemented during our interview with Ali would also help with this problem since it would help the robot from coming too close to humans unexpectedly. Furthermore, we planned to incorporate smoother motions into the robot while developing our code to prevent the robot from having any sudden or jerky movements that would make visitors uncomfortable.

Article Research

Navigation Method Evaluation

In order to find different ways that we can implement navigation methods in a human environment, we read an academic paper titled "A Study in Zucker: Insights on Interactions between Humans and Small Service Robots" by Alex Day and Ioannis Karamouzas which explained different social navigation robots and their merits. The most important aspect our team learned from this paper regarding the navigation system is the safety of humans. As stated in the paper, " robots should aim to maximize human safety and efficiency even at the cost of their own efficiency" [4]. Even if the robot does not complete its task in a fully efficient manner, if it avoided a human collision or prevented a situation in which someone would end up hurt, that is still much better than the alternative. If people do not feel safe around the robot, and the robot attempts to complete its tasks at the expense of human lives and their well-being, then there is no point in implementing the robot in the first place.

This paper compared two navigation methods: an anticipatory method that predicts human trajectories using probability mapping and deep learning, and a non-anticipatory method where a robot follows a set path. After conducting tests comparing the efficiency between the two, Day and Karamouzas found that the non-anticipatory method was both safer and more efficient, especially when the paths of people intersected with that of the robot. Taking into consideration

its much simpler integration and safer results, as well as its suitability for a museum environment where visitors may often cross paths with the robot, our team decided to implement a non-anticipatory method. We chose to use a colored tape line for navigation where the robot would only follow the line. This would not only avoid the unnecessary need for more complex code and hardware but most importantly improve user safety.

Design Features

Line tracking

One core feature of our robot is the ability to navigate to certain locations, specifically art exhibits. We decided that the most straightforward way to implement this feature would be through a line-following robot. One benefit of this is that it would allow the user of the robot to easily modify where they would like the robot to go by simply moving where the 'line' is. To implement this feature, we utilized the TCRT5000 sensor. This sensor is an Infrared, or IR, sensor, meaning that it is able to detect how far away something is. Although an IR is typically used to measure distance, we are able to utilize the fact that different colors reflect different amounts of light in order to follow lines. For example, since the colors black and white reflect less and more light respectively, the module will perceive them as being at different distances, even if they are right next to each other.

First, before attaching the sensors, we tested the sensors and adjusted their sensitivity accordingly. Then, we made sure to update our code to detect our chosen tape color. We placed a total of three sensors on the underside of the robot to detect the left, right, and center, as seen in **Figure 2.**

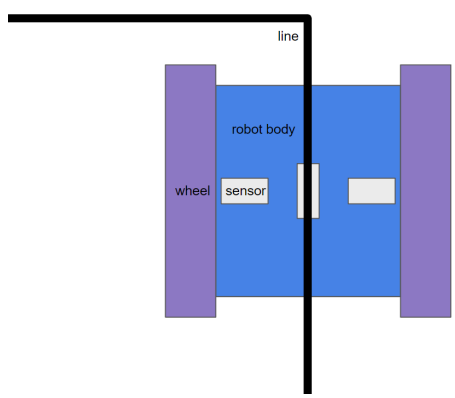

Figure 2: A diagram showcasing the underside of our robot
Credit: Ethan Ko

This placement allows for each sensor to focus on a different task. If the center sensor senses a line, the robot will proceed forward. However, if one of the other sensors detects a line, as seen in **Figure 3**, the robot will immediately rotate in that direction, then proceed forward again.

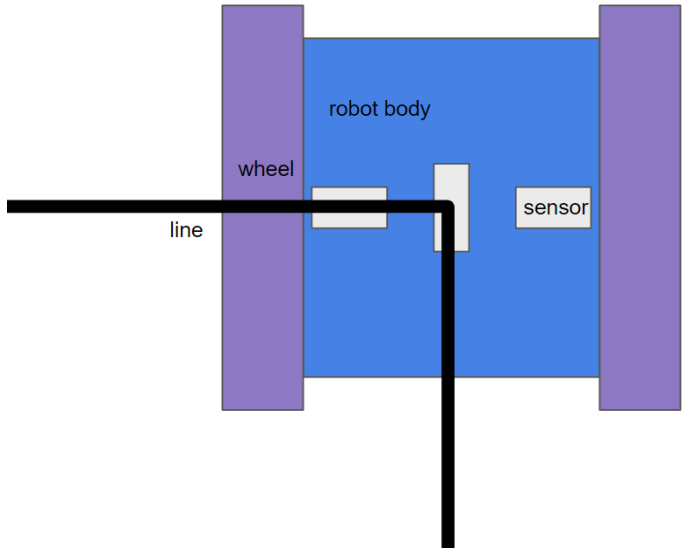

Figure 3: A diagram showcasing the underside of our robot as the line turns left
Credit: Ethan Ko

We chose to have the robot rotate instead of a gradual or moving turn as this would assist the robot in staying on track with the line. We made sure to communicate with the object detection system, meaning that the robot will only move when an object is not detected. When none of the sensors detected a line, it meant that the robot arrived at its destination. Then, it would call the information feature of the robot.

Object Detection

Another core feature of our robot is to detect objects and stop when it comes across them as the safety of visitors is of utmost importance during the operation of our robot. Since this is just a prototype version that we want to first get done then iterate on, we decided that a simple infrared sensor was the best option for object detection. However, we decided that, instead of getting another TCRT5000 sensor, we would get a Sharp brand sensor because it would make it easier to differentiate between line-following sensors and the object detection sensor if we did so. We specifically went with the Sharp GP2Y0A21YK0F. The sensor can detect ranges from 10-80 cm, which is over 30 inches. This meets our requirement of being able to detect objects within 1 foot of the robot since it is greater than the 5-inch lower bound of the sensor and less than the 30-inch upper bound. As can be seen in **Figure 4**, the sensor will be placed at the front of the robot.

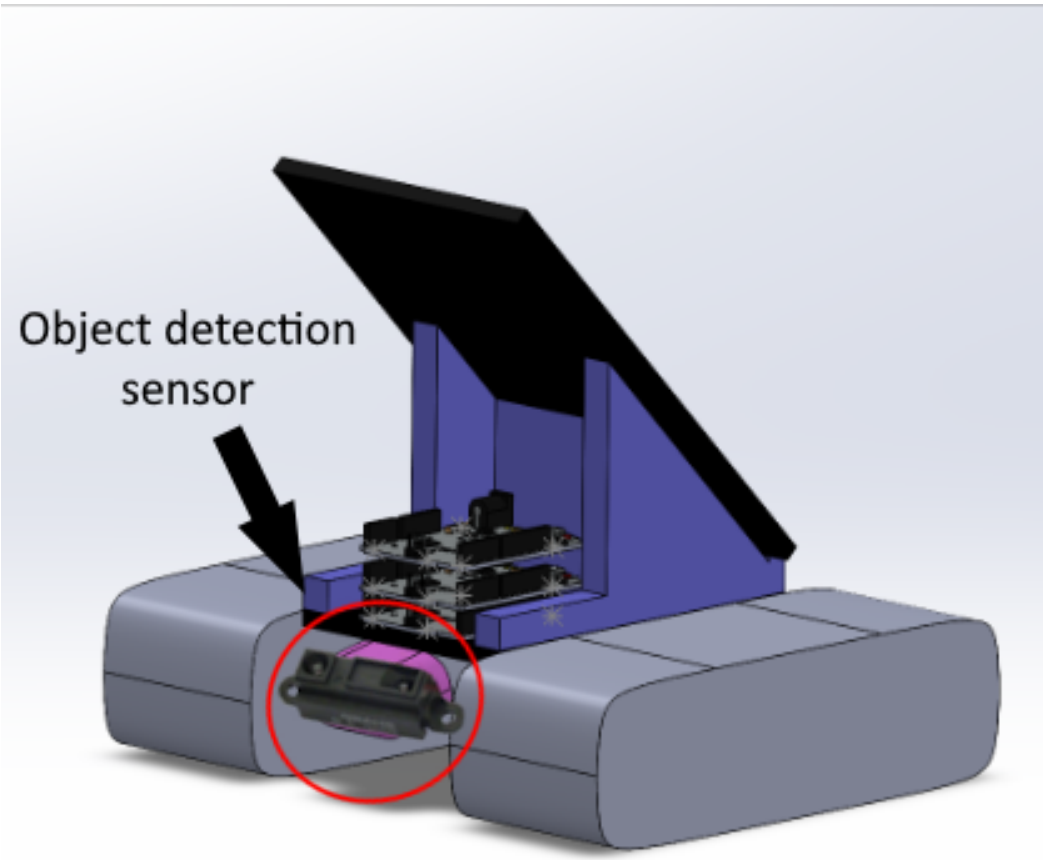

Figure 4: Guide Robot CAD design with object detection sensor
Credit: Grant Vermeesch

We chose to place the sensor at the front of the robot base in the direction that the robot was moving. This made the most sense because if something was in the path of the robot as it was following the line, then it would stop if the sensor detected an object.

Additionally, the object detection algorithm will communicate with the line following code and will take higher priority. So, the code will continuously check whether or not there is an object detected within range of the object detection sensor. While no object is detected, the robot will check each line following the sensor and follow the algorithm as described in the previous section.

Web-app interface

The key feature of our robot is the web-app interface, which facilitates human-robot interactions with our Museum Guide Bot. The web app features 5 intractable buttons, including “Language”, “Exhibits”, “Accessibility”, “About”, and “Connect Bluetooth”. This can all be viewed in a friendly to use interface when the user approaches the robot (see **Figure 5** for reference).

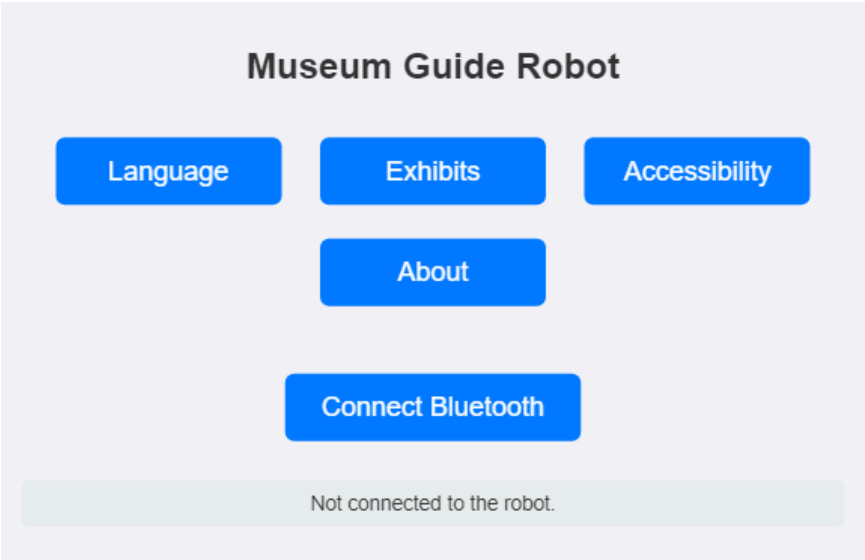

Figure 5: Guide Robot Web Interface
Credit: Haotian Li

The language option allows the visitor to choose their language of preference, this will change all the display and guide language to the selected language. Although this feature is not currently available, we hope to add it later. The exhibits option allows the visitor to view a list that includes all the exhibits in the museum. For our table-top demonstration, there is only one exhibit option available. Upon clicking an exhibit, the guide robot will follow a path that leads to the exhibit and explain the exhibit in detail. The accessibility tab will have special features for those with hearing and vision impairment. For those who have trouble with hearing, the guide will switch to text mode which will display text instead of audio when explaining the exhibits. For those with a vision impairment, the guide robot will use voice commands instead for exhibits and play a sound cue that the visitor can follow. The above option includes information about the guide bot, museum, and miscellaneous information that may be useful to the visitor.

By default, the web interface will be connected to the guide bot via Bluetooth through an Arduino. In case of any disconnection, the visitor can press the “Connect Bluetooth” option which will prompt the user with a list of Bluetooth devices. The user can then select to connect to the museum guide bot.

Drive-train

One of the basic functions of our robot was the ability to maneuver and lead visitors through the museum. We decided on a tank base with each side controlled by one motor leading into a motor controller connecting to our breadboard and the battery used to power them. We chose the tank drive style with the goal of close-quarters maneuverability and ease of coding in mind. We did run into some issues along the way such as the wiring which was slightly tricky seeing how we had to run it along with the other sensors on the bottom of the robot without anything getting

caught in any moving parts. Another slight problem was the tension of the tank treads which we could not get to be even so one was slightly looser than the other which caused a slight loss in the efficiency of driving. Even with these slight issues we ran into our drivetrain works just as intended with little issues.

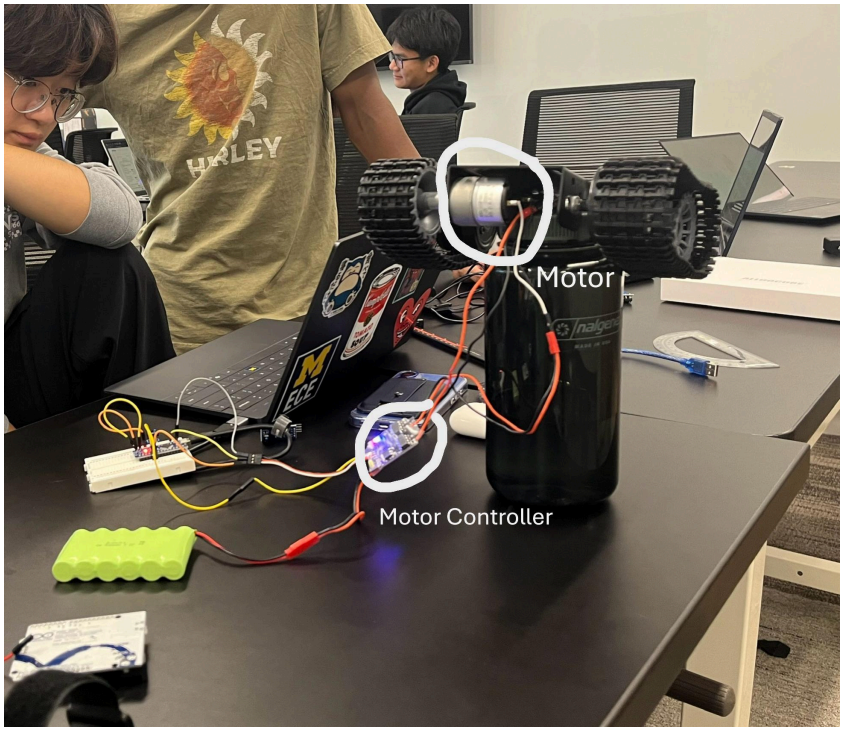

Figure 6: Guide Robot Table Top Model Drivetrain
Credit: Grant Vermeesch

Power Distribution

Perhaps the most important part of the entire robot was figuring out how to power it and distribute the proper voltages to each component. If we didn't figure out how the power distribution worked, then either the components would not get the power they need to function or they would get too much power and would get burnt and unusable. This was all the more important since we had to power the Arduino, the drive train, and all the sensors as well as the Bluetooth module. Since we had so many components, there weren't enough inputs and outputs for voltage regulation on the Arduino board itself. So, we decided to use a breadboard so that we could properly give power to all of the components. One part of the power distribution between the Arduino and Bluetooth module using the breadboard can be seen below in **Figure 7**. The 5V output from the Arduino goes to the power line of the breadboard and the Bluetooth module is also connected to the power terminal of the breadboard so it gets the power that is supplied to the Arduino from our main battery.

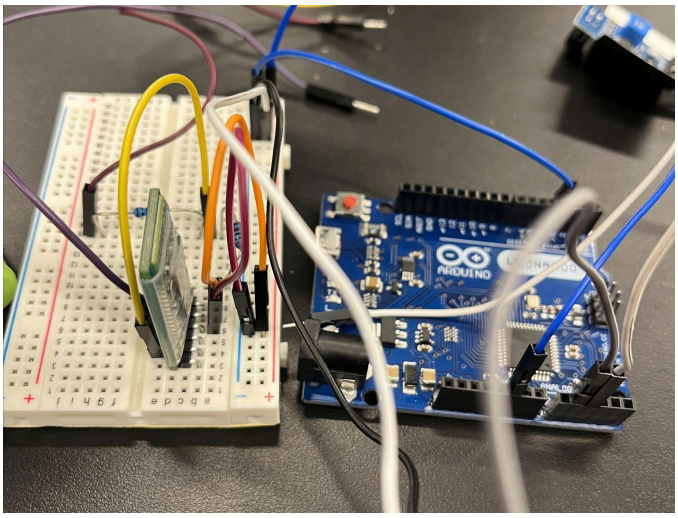

Figure 7: Arduino distributing power to bluetooth module through breadboard
Credit: Ethan Ko

Another part of the breadboard and arduino connection can be shown below in **Figure 8**, which shows how information is both received and transmitted between the arduino and bluetooth module.

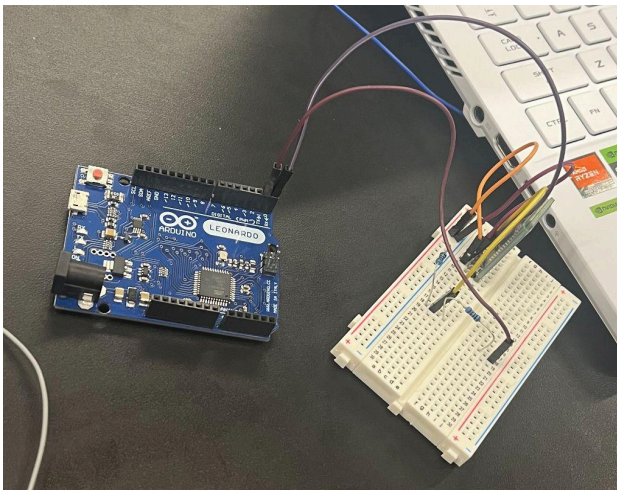

Figure 8: Arduino receiving and transmitting information to bluetooth module
Credit: Ethan Ko

In **Figure 8**, the 'RX' and 'TX' pins on the Arduino are how it both receives and transmits information to another component. It can be seen that the two pins are connected to the corresponding transmitting and receiving pins for the Bluetooth module on the breadboard. There is also some more complex wiring that includes things like resistors, which are gray wires with a blue coil in the middle, that are needed for the Bluetooth module to properly function.

CAD Base/Stand

For our model, we decided to CAD the initial vision for the project once we got past the drawing phase. We decided to use Solidworks to be the most accessible for our needs in this project. We took measurements of all the pieces that we intended to use and built a rough design to start with and use to show others our plan for the project. We created a base piece that is attached to our mounting plate for the tank treads. It was made with the intent to securely hold our tablet at an angle for the easiest access to the screen and options.

We flushed out the design of the model over the first couple of dedicated labs until we came up with a design that we thought would suit our needs. The first design can be seen in **Figure 9** below. It needed a couple of key features, one of the main ones is we had to raise it roughly 0.5 in off of the base plate since the treads were higher than the mounting plate. Another is it needs a lip to stop the tablet from sliding off of it while the robot drives around and is in use. We 3D printed our base and mounted it on the base and it fit decently with enough room for the Arduino and the battery.

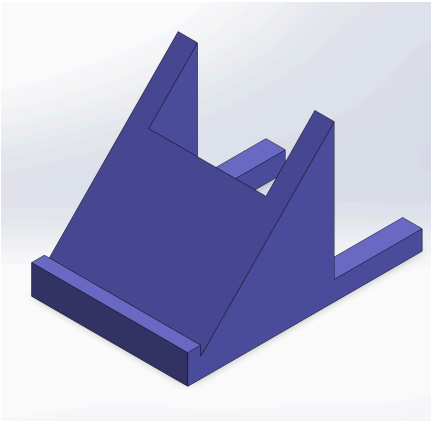

Figure 9: Guide Robot Table Top Base #1
Credit: Grant Vermeesch

However even with the success of the first version we decided we needed more room as we expanded our breadboard and Arduino setup. We extended our base platform out to the side to fit in more housing for the electronics and wiring as we expanded which can be seen in **Figure 10** on page 15. We decided to make two separate compartments overtop of the treads to try and maximize room while minimizing the total profile of the robot.

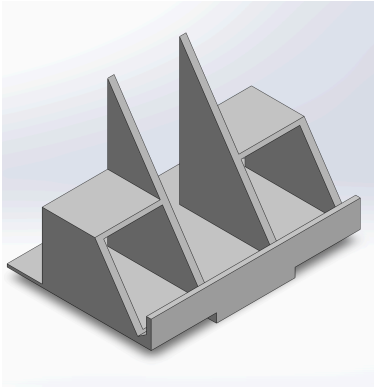

Figure 10: Guide Robot Table Top Base #2
Credit: Grant Vermeesch

Robot Information and Knowledge

As the purpose of this robot is to be a museum guide, it is important for it to be able to retell information to visitors. We decided that the best way to do this was through text on the screen. In conjunction with the line-following system, when the robot detects that it has reached its destination, it will output a predetermined set of information based on the exhibit. Rather than AI-generated information, we decided that information would be accurate if it came from qualified individuals more knowledgeable about the exhibit. This would also allow for the robot to output information as similar as possible to human guides, ensuring that nobody loses out on any important information.

Testing and Results

Line Following

In order to test the code of our Line Following feature, we created various 'test cases' to rigorously check its validity. This means that we created virtual simulations of plausible scenarios to see how the code would react. However, this feature does not rely only on code. In order to test the physical components, we utilized a combination of multimeters and components on the sensors. We utilized the multimeter to see if each sensor was receiving power and if they actually recognized lines. Another way that we were able to check this was through the sensors themselves. The TCRT5000 sensors come with a built-in light. This light will glow when the sensor detects something. Through this, we were able to check the physical components.

Object Detection

Similar to how we checked the Line Following feature, we had to test both the code and physical components. Since the Object Detection feature also utilized an IR, we were able to use similar strategies to test this feature. However, we went with a Sharp branded IR sensor rather than the

TCRT5000. One issue that came with this new IR sensor was that it didn't come with the built-in light that the TCRT5000 did. This limited our options to test the physical components to multimeters and check the voltage. However, we still tested the sensing ability of the Sharp sensor by having our code output the values that the sensor wrote. Using the output values, we were able to see the distances of objects from the sensor that were being detected and tuned them to fit our needs. After some continuous testing, we decided that a distance of around 1 foot from the sensor was the best distance for object detection. One drawback of this technology is that, when objects are 1 foot away, there is not as much reflected light from objects that reach the sensor. This could result in cases where some darker objects or people wearing darker clothes could still not be detected by the sensor even if they are 1 foot away. In these cases, it will likely take longer for the robot to detect the object and stop less than 1 foot from it. This is something that we would fine-tune further in future iterations.

Conclusion and Next Steps

In order to combat the issue of dwindling human resources in museums, our team decided to devote ourselves to developing a robot museum guide prototype. We conducted interviews with avid museum-goers and robotic experts and came to the conclusion that a line-following robot would be most efficient for our purposes, allowing museum workers to easily change the robot's path. We also learned that we should minimize the jerky movements of the robot in order to ensure that the robot appears more friendly to potential visitors.

When ideating our project, we set out to create a large guide robot that contained a variety of accessibility features and foreign language support. We hoped that it would have the capability of navigating to as many museum exhibits as needed with the goal of creating a better experience for all museum guests. However, instead of creating the large full-size version that we would implement in the real world, we decided to build a smaller desktop version. Throughout the timeline of this proof of concept project development, we were able to create a working prototype robot that was able to follow lines, detect obstacles, and provide information. Nevertheless, there are still many steps to be made in our design and testing. Next steps include expanding accessibility options on the guide robot. Some examples of this may include foreign language support, speech support for individuals with hearing impairments, and different movement modes for people with movement disorders. We would also implement a speaker system as it would likely improve the robot's overall use and efficiency. These would hopefully allow our robot to be usable by all guests, enhancing their museum experience in the process.

References

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Appendix

Interview with Zulfeqar Ali over phone call relevant quotes: October 20, 2024

- “You can have a camera sense, not only the line, but also people around the robot so it doesn’t accidentally dash into anything”
- [When talking about what sensors to use] “Radar can also be done. It is less expensive than LiDAR, but they are both more expensive than just basic sensors.”